

Escolha Múltipla

3,5 / 3,6

1

$$R \Big|_{T=20^{\circ}\text{C}} = 6,5 \text{ k}\Omega$$

$$\alpha_{20} = ?$$

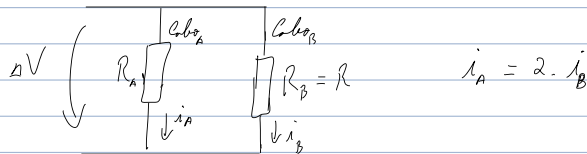
$$R \Big|_{T=60^{\circ}\text{C}} = 8 \text{ k}\Omega$$

$$R = R_{20} \cdot (1 + \alpha_{20} (T - 20)) \Leftrightarrow 8 \text{ k} = 6,5 \text{ k} \cdot (1 + \alpha_{20} (60 - 20))$$

$$\Leftrightarrow 5,77 \text{ m } ^{\circ}\text{C}^{-1}$$

B

2



$$\Delta V = R \cdot i$$

$$\begin{cases} \Delta V = R_A \cdot i_A \\ \Delta V = R_B \cdot i_B \end{cases} \Leftrightarrow \begin{cases} \Delta V = R_A \cdot 2 \cdot i_B \\ \Delta V = R \cdot i_B \end{cases} \Rightarrow R_A \cdot 2 \cdot i_B = R \cdot i_B \Rightarrow R_A = R/2$$

C

3

Dois fios de cobre ; $V_1 = V_2 = V$; $L_2 = 3,2 \cdot L_1$; $R_2 / R_1 = ?$

$$R = \rho \cdot \frac{L}{A}$$

$$A_1 = \frac{V}{L_1} ; A_2 = \frac{V}{L_2}$$

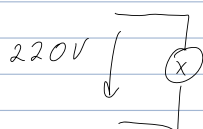
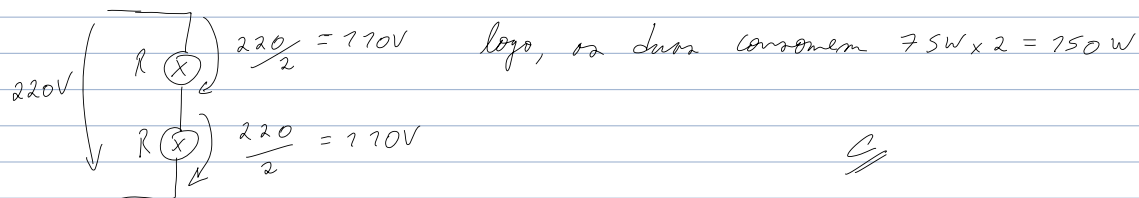
$$\begin{cases} R_1 = \rho_{cu} \cdot \frac{L_1}{\frac{V}{L_1}} = \rho_{cu} \cdot \frac{L_1^2}{V} \\ R_2 = \rho_{cu} \cdot \frac{L_2}{\frac{V}{L_2}} = \rho_{cu} \cdot \frac{L_2^2}{V} \end{cases} \Leftrightarrow \frac{R_2}{R_1} = \frac{\rho_{cu} \cdot \frac{L_2^2}{V}}{\rho_{cu} \cdot \frac{L_1^2}{V}} = \left(\frac{L_2}{L_1} \right)^2 \Rightarrow$$

$$\Rightarrow \frac{R_2}{R_1} = \left(\frac{3,2 \cdot L_1}{L_1} \right)^2 = 3,44 \text{ } \Omega / \Omega$$

E

4) Quando é aplicado 220V ela tem uma potência de 75W

Se ligar 2 em série:



5)

100V $P = 1000 \text{ W} \Leftrightarrow P = V \cdot i = V \cdot \frac{V}{R} = \frac{V^2}{R} \Leftrightarrow R = \frac{100^2}{1000} = 10 \Omega$

120V $= 120V \text{ } \quad P = \frac{V^2}{R} = \frac{120^2}{3 \times 10} = 480 \text{ W}$

Problemas

7)

Fio de cobre; $T_0 = 12^\circ\text{C}$; $T_1 = ?$ para que $R_1 = 1,1 \cdot R_0$

$$R = R_{20} \cdot (1 + \alpha_{20} \cdot (T - 20))$$

$$R_0 = R_{20} \cdot (1 + 0,0039 \cdot (12 - 20)) = 0,969 \cdot R_{20}$$

$$R_1 = 1,1 \cdot (0,969 \cdot R_{20}) = \cancel{R_{20}} \cdot (1 + 0,0039 \cdot (T_1 - 20)) \Leftrightarrow$$

$$\Rightarrow T_1 = 36,9^\circ \text{C}$$

$$T_1 - T_0 = 36,9 - 12 = 24,9^\circ \text{C}$$

[2]

$$\text{Fio 1} \rightarrow L_1 = 1 \text{ m}; R_1 = 0,3 \Omega; \text{volume} = V$$

$$\text{Fio 2} \rightarrow L_2 = 2 \text{ m}; R_2 = ?; \text{volume} = V$$

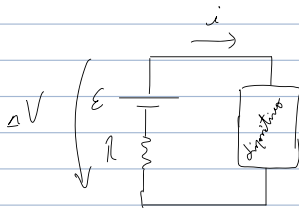
$$R = \rho \cdot \frac{L}{A} = \rho \cdot \frac{L^2}{V}$$

$$R_1 = 0,3 = \rho \cdot \frac{1^2}{V} \Rightarrow \frac{\rho}{V} = 0,3$$

$$R_2 = \rho \cdot \frac{2^2}{V} = 0,3 \times 2^2 = 1,2 \Omega$$

[3]

$$\Delta V_{\text{gerador}} = \mathcal{E} - r \cdot I$$



$$\Delta V = \mathcal{E} + R \cdot i$$

①:

$$\Delta V_1 = 3 \text{ V}$$

$$i_1 = -4 \text{ A}$$

Operação de receptor

②:

$$\Delta V_2 = 12 \text{ V}$$

$$i_2 = 2 \text{ A}$$

Operação de gerador

$$\begin{aligned} \text{①: } \begin{cases} 3 = \mathcal{E} + R \cdot (-4) \\ 12 = \mathcal{E} + R \cdot 2 \end{cases} &\Rightarrow \begin{cases} 4 \cdot R = \mathcal{E} - 3 \\ \mathcal{E} = 12 - 2 \cdot R \end{cases} \Rightarrow \begin{cases} 4 \cdot R = 12 - 2 \cdot R - 3 \\ \text{---} \end{cases} \Rightarrow \\ &\Rightarrow \begin{cases} R = 1,5 \Omega \\ \mathcal{E} = 9 \text{ V} \end{cases} \end{aligned}$$

[4]

$$\mathcal{E} = 1,5 \text{ V}$$

$$R_{\text{int}} = 0,2 \Omega$$

A corrente máxima ocorre quando a pilha é ligada em curto-circuito, $R_{\text{exterior}} = 0 \Omega$:

$$\Delta V = R \cdot i \rightarrow i_{\text{máx}} = \frac{1,5}{0,2} = 7,5 \text{ A}$$

$$P \Big|_{i=7,5 \text{ A}} = V \cdot i = i \cdot R \cdot i = R \cdot i^2 = 0,2 \cdot 7,5^2 = 11,25 \text{ W}$$

5

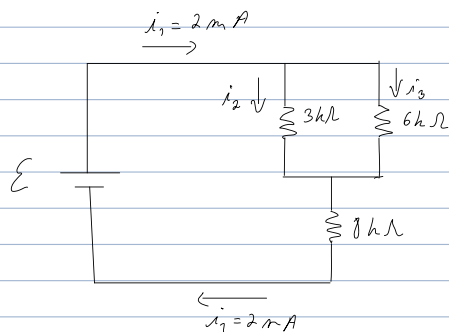
$$P = 60 \text{ W}; \Delta V = 230 \text{ V}; R_{20^\circ\text{C}} = 65 \Omega$$

$$R = R_{20} \cdot (1 + \alpha_{20} \cdot (T - 20)) \Leftrightarrow R = 65 \cdot (1 + 0,0045 \cdot (T - 20)) \Leftrightarrow$$

$$\Rightarrow T = \frac{\left(\frac{R}{65} - 1\right)}{0,0045} + 20 = 3,42 \cdot 10^3 - 202,2$$

Assim, a resistência varia linearmente com a temperatura, e a temperatura da lâmpada agora é muito maior que 20°C .

6



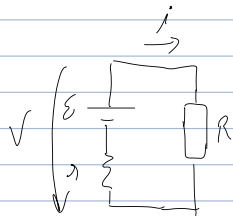
$$\left\{ \begin{array}{l} i_1 = i_2 + i_3 \\ \Delta V_{3k\Omega} = i_2 \cdot 3k \\ \Delta V_{6k\Omega} = i_3 \cdot 6k \\ \Delta V_{8k\Omega} = 0,002 \cdot 8k = 16 \text{ V} \\ \Delta V_{3k\Omega} = \Delta V_{6k\Omega} \\ E = \Delta V_{3k\Omega} + \Delta V_{8k\Omega} \end{array} \right. \Leftrightarrow \left\{ \begin{array}{l} i_2 \cdot 3k = i_3 \cdot 6k \Leftrightarrow i_2 = 2 \cdot i_3 \end{array} \right. \Rightarrow$$

$$\left\{ \begin{array}{l} 0,002 = 2 \cdot i_3 + i_3 \Leftrightarrow i_3 = 0,002/3 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} i_2 = i_1 - i_3 = 0,002 - \frac{0,002}{3} = \frac{0,004}{3} \end{array} \right.$$

$$\Rightarrow \left\{ \begin{array}{l} \Delta V_{3k\Omega} = \frac{0,004}{3} \cdot 3k = 4 \text{ V} \\ \Delta V_{6k\Omega} = \frac{0,002}{3} \cdot 6k = 4 \text{ V} \end{array} \right. \Rightarrow E = \Delta V_{3k\Omega} + \Delta V_{8k\Omega} = 4 + 16 = 20 \text{ V}$$

7

a)

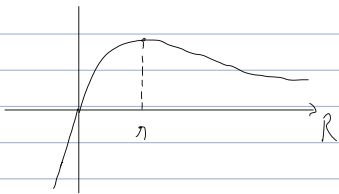


$$P_R = R \cdot i^2 = R \cdot \left(\frac{E}{R+r} \right)^2 =$$

$$= \frac{R \cdot E^2}{R^2 + r^2 + 2 \cdot R \cdot r} = \frac{E^2}{\frac{r^2}{R} + 2r + R}$$

Para encontrar pontos máximos e mínimos:

$$\frac{d}{dR} \left(\frac{r^2}{R} + 2r + R \right) = 0 \Leftrightarrow -\frac{r^2}{R^2} + 1 = 0 \Leftrightarrow r = R \text{ é a única solução.}$$



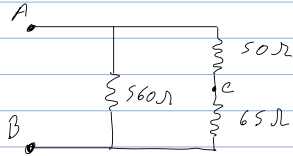
b)

$$P = \frac{E^2}{\frac{r^2}{R} + 2r + R}$$

$$P_{\max} \Big|_{R=r} = \frac{E^2}{\frac{r^2}{r} + 2r + r} = \frac{E^2}{4r}$$

8

$R_{AB} :$



$$R_{AB} = 560 \parallel (50 + 65) = \frac{560 \cdot 115}{560 + 115} = 95,4 \, \Omega$$

$$R_{AC} = 50 \parallel (560 + 65) = \frac{50 \cdot 625}{50 + 625} = 46,3 \, \Omega$$

$$R_{BC} = 65 \parallel (560 + 50) = \frac{65 \cdot 610}{65 + 610} = 58,7 \, \Omega$$

No circuito em estrela :

$$\begin{cases} R_{AB} = R_1 + R_2 = 95,4 \\ R_{AC} = R_1 + R_3 = 46,3 \\ R_{BC} = R_2 + R_3 = 58,7 \end{cases} \Leftrightarrow \begin{cases} R_1 = 95,4 - R_2 \\ R_3 = 46,3 - 95,4 + R_2 \\ R_2 = 58,7 - 46,3 + 95,4 - R_2 \end{cases} \Rightarrow$$

$$R_1 = 41,5 \, \Omega$$

$$\Rightarrow R_3 = 4,8 \, \Omega$$

$$R_2 = 53,9 \, \Omega$$

[9]

Fio cobre; diâmetro = 1,29 mm; $i_{\text{máx}} = 6 \text{ A}$

a)

$$T = 20^\circ\text{C} \quad \Delta V \Big|_{d=40\text{ m}} = ?$$

$$R_{20} = ? = \rho_{\text{Cu}} \cdot \frac{d}{A} = 17 \times 10^{-9} \cdot \frac{40}{\pi \cdot \left(\frac{1,29 \times 10^{-3}}{2}\right)^2} = 520 \text{ m}\Omega$$

$$\Delta V = 6 \times 0,520 = 3,12 \text{ V}$$

b) $P_{\text{pr}} = i^2 \cdot R = 6^2 \times 0,520 = 18,7 \text{ W}$

[10]

fio de níquel - cromo; diâmetro = 1,8 mm; 8 g de vapor por segundo;
 $V = 220 \text{ V}$; $d = ?$; $2257,2 \text{ J/g}$; $T = 100^\circ\text{C}$

$$R_{20} = \frac{1 \times 10^{-6} \cdot d}{\pi \cdot (0,9 \times 10^{-3})^2} = 393 \cdot d \text{ m}\Omega$$

$$R_{100} = 393 \cdot d \cdot 10^{-3} \cdot (1 + 0,4 \times 10^{-3} (100 - 20)) = 406 \cdot d \text{ m}\Omega$$

$$P = 2257,2 \times 8 = 18,06 \text{ kJ/s} = 18,06 \text{ kW} = \frac{V^2}{R_{100}}$$

$$R_{100} = \frac{220^2}{18,06 \times 10^3} = 406 \cdot d \times 10^{-3} \Rightarrow d = 6,6 \text{ m}$$

[11]

Fio cobre:

$$R_{20} = 17 \times 10^{-9} \cdot \frac{0,32}{\pi \cdot \left(\frac{0,7 \times 10^{-3}}{2}\right)^2} = 693 \text{ m}\Omega$$

$$R_{\text{cu}} = 693 \times 10^{-3} \cdot (1 + 0,0039 \cdot (T - 20))$$

$$= 0,693 \cdot (1 + 0,0039 \cdot T - 0,078)$$

$$= 0,6386 + 0,0027 \cdot T$$

Fio de Tungstênio:

$$R_{20} = 55 \times 10^{-9} \cdot \frac{0,7}{\pi \cdot \left(\frac{0,7 \times 10^{-3}}{2}\right)^2} = 700 \, \mu\Omega$$

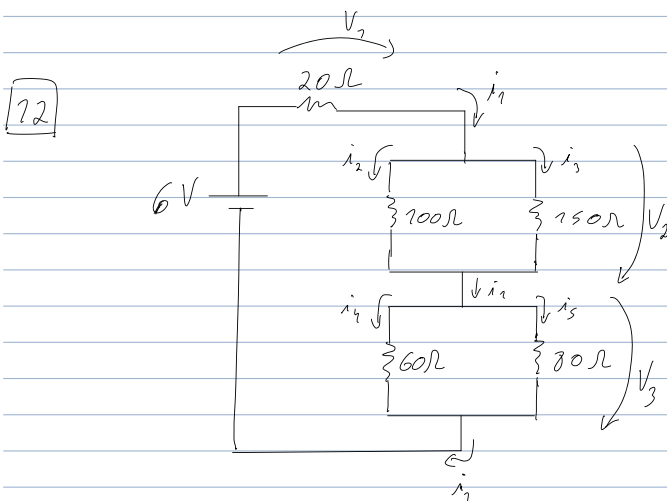
$$R_w = 700 \times 10^{-3} \cdot (1 + 0,0045 \cdot (T - 20))$$

$$= 0,7 \cdot (1 + 0,0045 \cdot T - 0,09)$$

$$= 0,6373 + 0,00315 \cdot T$$

$$R_{cu} = R_w \Leftrightarrow 0,6386 + 0,0027 \cdot T = 0,6373 + 0,00315 \cdot T \Leftrightarrow$$

$$\Leftrightarrow T = 2,9 \, ^\circ\text{C}$$



$$\begin{cases} i_1 = i_2 + i_3 \\ i_1 = i_4 + i_5 \\ 6 = V_1 + V_2 + V_3 \end{cases} \Leftrightarrow \begin{cases} \frac{V_1}{20} = \frac{V_2}{100} + \frac{V_2}{150} \\ \frac{V_1}{20} = \frac{V_3}{60} + \frac{V_3}{80} \end{cases} \Leftrightarrow \begin{cases} V_1 = \frac{2 \cdot V_2}{10} + \frac{2 \cdot V_2}{15} \\ V_1 = \frac{2 \cdot V_3}{6} + \frac{2 \cdot V_3}{8} \end{cases} \S$$

$$\begin{cases} V_1 = \frac{7}{3} \cdot V_2 \\ V_1 = \frac{7}{12} \cdot V_3 \\ 6 = V_1 + 3 \cdot V_1 + \frac{12}{7} \cdot V_1 \end{cases} \Leftrightarrow \begin{cases} V_2 = 3,15 \text{ V} \\ V_3 = 3,8 \text{ V} \\ V_1 = 3,05 \text{ V} \end{cases}$$

$$P_{20\Omega} = \frac{1,05^2}{20} = 55,1 \text{ mW}$$

$$P_{100\Omega} = \frac{3,15^2}{100} = 99,2 \text{ mW}$$

$$P_{150\Omega} = \frac{3,15^2}{150} = 66,2 \text{ mW}$$

$$P_{60\Omega} = \frac{1,8^2}{60} = 54,0 \text{ mW}$$

$$P_{80\Omega} = \frac{1,8^2}{80} = 40,5 \text{ mW}$$

$$P_{\text{Total}} = 375 \text{ mW}$$

$$R_{eq} = 20 + (100 \parallel 150) + (60 \parallel 80) = 20 + \frac{100 \times 150}{100 + 150} + \frac{60 \times 80}{60 + 80} = 114,29 \Omega$$

$$P = \frac{6^2}{114,29} = 375 \text{ mW}$$